

Cyclic Group

Definition:

A group $(G,*)$ is said to be cyclic if there exists $a \in G$ such that any $x \in G$ can be written as either $x = a^n$ or $x = na$, where n is some integer. Here the element ' a ' is called the 'generator' of the cyclic group G . That is the cyclic group generated by ' a ' and we denote it by $G = \langle a \rangle$

Theorem 1:

Every cyclic group is abelian.

Proof:

Let $(G,*)$ be a cyclic group generated by an element $a \in G$. i.e $G = \langle a \rangle$. Then for any two elements $x, y \in G$, we have $x = a^n, y = a^m$, where m, n are integers.

$$\begin{aligned}x * y &= a^n * a^m = a^{n+m} = a^{m+n} \\&= a^m * a^n \\&= y * x\end{aligned}$$

$(G,*)$ is abelian.

Theorem 2:

If 'a' is a generator of a cyclic group G , then a^{-1} is also a generator of G (or). If a is an element of group G , then $(a) = (a^{-1})$.

Proof. Let $G = (a)$ be a cyclic group generated by 'a'. Then $a^r \in G$, where r is some integer we can write $a^r = (a^{-1})^{-r}$, since $-r$ is also some integer. Each element of G is generated by a^{-1} .

a^{-1} is also a generator of G . $(G) = (a) = (a^{-1})$.

Theorem 3:

Prove that the group $G = \{1, -1, i, -i\}$ is cyclic and find its generators.

Solution. We can write $1 = (i)^4, -1 = (i)^2, i = (i)^1, -i = (i)^3$

i.e, All the elements of G can be expressed as integral powers of the element i.e.

G is a cyclic group generated by i . Since 'i' is the generator of G , $(i)^{-1}$ is also a generator of G . Hence G is a cyclic group and its generators are i and $(i)^{-1} = 1/i = -i$

Theorem 4:

Every subgroup of a cyclic group is cyclic

Theorem 5:

Let $(G,*)$ be a finite group generated by an element $a \in G$. If G is of order n , that is $O(G) = n$, then $a^n = e$ so that $G = \{a, a^2, \dots, a^n = e\}$. Further, n is the least positive integer for which $a^n = e$.

Example: Show that every group of order 3 is cyclic.

Solution:

Since G is of order 3, it must have two distinct elements a, b apart from the identity element e .

Since G is closed under the operation $*$,

$$a * b \in G$$

$$\therefore a * b = a \text{ or } a * b = b \text{ or } a * b = e$$

If $a * b = a$, $a * b = a * e$

$\therefore b = e$, by cancellation law

If $a * b = b$, $a * b = e * b$

$\therefore a = e$, by cancellation law

But a and b are not equal to e .

$\therefore a * b = e$ (1)

Again by closure law, $a^2 \in G$

$\therefore a^2 = a$ or $a^2 = b$ or $a^2 = e$

If $a^2 = a$ or $a * a = a * e$, then $a = e$, which is not true

If $a^2 = e$, then $a^2 = a * b$, by (1)

$\therefore a = b$, which is not true.

$\therefore a^2 = b$ (2)

Also $a^3 = a * a^2 = a * b$, by (2)

$= e$, by (1)

Hence $G = \{a, a^2, a^3 (=e)\}$ is a cyclic group with generator a .

Coset

Definition: If G is a group and H is a subgroup of G , then,

- for any $a \in G$, the set $a * H$ defined by $a * H = \{a * h \mid h \in H\}$ is called the **Left coset** of H in G determined by the element $a \in G$.
- for any $a \in G$, the set $H * a$ defined by $H * a = \{h * a \mid h \in H\}$ is called the **Right coset** of H in G determined by the element $a \in G$.

The element $a \in G$ is called the representative element of the left coset $a * H$ and right coset $H * a$

Note The right or left coset of H in G is not empty.

Since $e \in H$, $e * H = H = H * e$, H itself is right as well as left coset. $H * a$, $a * H$ are also subsets of G .

If G is abelian, then $a * H = H * a$ In all the subsequence discussion in this chapter, aH means $a * H$ and Ha means $H * a$

Example.1

Let $G = \{1, a, a^2, a^3\}$ be a group and $H = \{1, a^2\}$ is a subgroup of G under multiplication. Find all the cosets of H . where $a^4 = 1$

Solution:

Let us find the right cosets of H in G .

$$H1 = \{1, a^2\} = H$$

$$Ha = \{a, a^3\},$$

$$H a^2 = \{a^2, a^4\} = \{a^2, 1\} = H$$

$$H a^3 = \{a^3, a^5\} = \{a^3, a\} = Ha$$

$$H.1 = H = H a^2 = \{1, a^2\} \text{ and } Ha = H a^3 = \{a, a^3\}$$

are two distinct right cosets of H in G . Similarly we can find the left cosets of H in G .

Example.2 Find the left cosets of $\{[0], [2]\}$ in the group $(Z_4, +_4)$.

Solution Let $Z_4 = \{[0],[1],[2],[3]\}$ be a group and $H = \{[0],[2]\}$ be a subgroup of Z_4 under $+_4$ (addition mode 4).

The left cosets of H are

$$[0] + H = \{[0],[2]\} = H ;$$

$$[1] + H = \{[1],[3]\} ;$$

$$[2] + H = \{[2],[4]\} = \{[2],[0]\} = \{[0],[2]\} = H$$

$$[3] + H = \{[3],[5]\} = \{[3],[1]\} = \{[1],[3]\} = [1] + H.$$

$[0] + H = [2] + H = H$ and $[1] + H = [3] + H$ are the two distinct left cosets of H in Z_4 .

Note: The union of all distinct left cosets of H in G is equal to G .
If H any subgroup of G and $a \in H$, then $Ha = H = aH$.

Theorem: Any two right cosets of H in G are either disjoint or identical.

Lagrange's theorem

The order of each subgroup of a finite group is a divisor of the order of the group.

Homomorphism and Isomorphism.

Homomorphism : Let $(G,*)$ and (H,Δ) be any two groups. A mapping $f: G \rightarrow H$ is said to be a homomorphism if $f(a*b) = f(a)\Delta f(b)$, for any $a, b \in G$.

Ex1: Let $g: (Z, +) \rightarrow (Z, +)$ defined by $g(x) = 2x$ for all $x \in Z$
Then for $x, y \in Z$

$$g(x + y) = 2(x + y) = 2x + 2y = g(x) + g(y)$$

$\therefore g$ is a homomorphism.

Ex2: Let $(R^+, \cdot)$ be a group of positive real numbers under multiplication. Let $(R, +)$ be the group of all real numbers under addition.

Define $f: R^+ \rightarrow R$ by $f(x) = \log_{10}^x$. The binary operation for the groups R^+ and R are $\cdot$ and $+$ respectively.

To prove: f is a homomorphism.

i.e. to show that $f(xy) = f(x) + f(y)$ for all $x, y \in R^+$

Now,

$$f(xy) = \log_{10}^{xy} = \log_{10}^x + \log_{10}^y = f(x) + f(y).$$

$\therefore f$ is a homomorphism.

Theorem: Consider the groups $(G_1, *)$ and $(G_2, \oplus)$ with identity elements e_1 and e_2 respectively. If $f: G_1 \rightarrow G_2$ is a group homomorphism, then prove that

a) $f(e_1) = e_2$

b) $f(a^{-1}) = [f(a)]^{-1}$

Proof: a) We have in G_2 ,

$$\begin{aligned} e_2 \oplus f(e_1) &= f(e_1) && \text{(since, } e_2 \text{ is identity in } G_2 \text{)} \\ &= f(e_1 * e_1) && \text{(since, } e_1 \text{ is identity in } G_1 \text{)} \\ &= f(e_1) \oplus f(e_1) && \text{(since } f \text{ is a homomorphism)} \\ e_2 &= f(e_1) && \text{(By right cancellation law)} \end{aligned}$$

b) For any $a \in G_1$, we have

$$f(a) \oplus f(a^{-1}) = f(a * a^{-1}) = f(e_1) = e_2$$

$$\text{and } f(a^{-1}) \oplus f(a) = f(a^{-1} * a) = f(e_1) = e_2$$

$\therefore f(a^{-1})$ is the inverse of $f(a)$ in G_2 i.e., $[f(a)]^{-1} = f(a^{-1})$

Theorem 2: Let f be a homomorphism from G to G' . Then $f(G)$ (the homomorphic image of G into G') is a subgroup of G'

Proof:

$$\text{Let } f(G) = \{f(x) \mid x \in G\}$$

Clearly $f(G)$ is a non-empty subset of G' . Now for any $a', b' \in f(G)$, we have $f(a) = a', f(b) = b'$, for $a, b \in G$.

$$\begin{aligned} a' * (b')^{-1} &= f(a) * f(b)^{-1} \\ &= f(a) * f(b^{-1}) \\ &= f(a * b^{-1}) \end{aligned}$$

But $a * b^{-1} \in G \Rightarrow f(a * b^{-1}) \in f(G) \Rightarrow a' * (b')^{-1} \in f(G)$
i.e., For $a', b' \in f(G) \Rightarrow a' * (b')^{-1} \in f(G)$
 $\Rightarrow f(G)$ is a subgroup of G' .

Isomorphism

Definition: A mapping f from a Group $(G, .)$ to a group $(G', *)$ is said to be an **Isomorphism** if

- f is a homomorphism i.e. $f(a . b) = f(a) * f(b)$, for any $a, b \in G$
- f is one-one. (injective) i.e. distinct elements of G have different f -image in G' .
- f is onto (surjective) i.e. every element of G' should be the f -image of some elements in G .

Ex 1: Let $f: (Z, +) \rightarrow (2Z, +)$ defined by $f(x) = 2x$ for all $x \in Z$ prove that f is an isomorphism.

Solution:

Given $f: (Z, +) \rightarrow (2Z, +)$

Defined by $f(x) = 2x$

Now $f(x + y) = 2(x + y) = 2x + 2y = f(x) + f(y)$

for all $x, y \in Z$

Therefore, f is a homomorphism.

Let $f(x) = f(y) \Rightarrow 2x = 2y \Rightarrow x = y$

$\therefore f$ is 1-1

Clearly f is onto.

Hence f is an isomorphism.

Ex 2. Let R be a group of all real numbers under addition and R^+ be a group of all positive real numbers under multiplication. Show that the mapping $f: R \rightarrow R^+$ defined by $f(x) = 2^x$ for all $x \in R$ is an isomorphism.

Solution: Homomorphism property is already shown.

Next, let us prove that f is a Bijection.

For any $a, b \in \mathbb{R}$, Let, $f(a) = f(b)$

$$\Rightarrow 2^a = 2^b$$

$$\Rightarrow a = b$$

$\therefore f$ is one-to-one.

Next, take any $c \in \mathbb{R}^+$.

Then $\log_2 c \in \mathbb{R}$ and $f(\log_2 c) = 2^{\log_2 c} = c$.

$\Rightarrow$ Every element in $\mathbb{R}^+$ has a pre image in $\mathbb{R}$.

i.e., f is onto.

$\therefore f$ is a bijection.

Hence, f is an isomorphism.

Theorem 3-2.1 Let $\langle S, * \rangle$, $\langle T, \Delta \rangle$, and $\langle V, \oplus \rangle$ be semigroups and $g: S \rightarrow T$ and $h: T \rightarrow V$ be semigroup homomorphisms. Then $(h \circ g): S \rightarrow V$ is a semigroup homomorphism from $\langle S, * \rangle$ to $\langle V, \oplus \rangle$.

PROOF Let $a, b \in S$. Then

$$\begin{aligned} (h \circ g)(a * b) &= h(g(a * b)) = h(g(a) \Delta g(b)) \\ &= h(g(a)) \oplus h(g(b)) \\ &= (h \circ g)(a) \oplus (h \circ g)(b) \end{aligned} \quad \text{//} //$$

Theorem 3-2.8 For any commutative monoid $\langle M, * \rangle$, the set of idempotent elements of M forms a submonoid.

PROOF Since the identity element $e \in M$ is idempotent, $e \in S$, where S is the set of idempotents of M . Let $a, b \in S$, so that

$$a * a = a \quad \text{and} \quad b * b = b$$

$$\text{Now } (a * b) * (a * b) = (a * b) * (b * a) \quad (\langle M, * \rangle \text{ is commutative})$$

$$= a * (b * b) * a$$

$$= a * b * a$$

$$= a * a * b = a * b$$

Hence $a * b \in S$ and $\langle S, * \rangle$ is a submonoid.

////

Theorem 3-5.3 A subset $S \neq \emptyset$ of G is a subgroup of $\langle G, * \rangle$ iff for any pair of elements $a, b \in S$, $a * b^{-1} \in S$.

PROOF Assuming that S is a subgroup, it is clear that if $a, b \in S$, then $b^{-1} \in S$ and $a * b^{-1} \in S$. To prove the converse, let us assume that $a, b \in S$ and $a * b^{-1} \in S$ for any pair a, b . Taking $b = a$, $a * a^{-1} = e \in S$. From $e, a, b \in S$, we have $e * a^{-1} = a^{-1} \in S$. Similarly, $b^{-1} \in S$. Finally, because a and b^{-1} are in S , we have $a * b \in S$. Hence $\langle S, * \rangle$ is a subgroup of $\langle G, * \rangle$. ////

Theorem 3-5.7 The order of a subgroup of a finite group divides the order of the group.

The number of left cosets of H in G is called the *index* of H in G . From Lagrange's theorem we have the index k of G given by

$$k = |G|/|H|$$

It immediately follows from Lagrange's theorem that any group of prime order has only trivial subgroups, while a group of order n which has divisors other than 1 and n has proper subgroups.

Recall the group $\langle D_4, \diamond \rangle$ of order 8 given in Table 3-5.7. $\langle D_4, \diamond \rangle$ has five different subgroups of order 2 and one subgroup of order 4. Similarly the symmetric group $\langle S_3, \diamond \rangle$ has subgroups of orders 2 and 3 besides the trivial subgroups.

For a group $\langle G, * \rangle$ we know that the cyclic group generated by any element of G must be a subgroup of G . Let $a \in G$ and the cyclic subgroup generated by a be denoted by H , that is, $H = \{a^i \mid i \in \mathbb{I}\}$. If H is of order m , then according to Lagrange's theorem m must divide n where n is the order of the group G . Hence from $a^m = e$ we also have $a^n = a^{m \cdot k} = (a^m)^k = e^k = e$. We therefore have the following corollary of Lagrange's theorem.

Corollary If $\langle G, * \rangle$ is a finite group of order n , then for any $a \in G$, we must have $a^n = e$, where e is the identity of the group G . ////

Thⁿ Let $(G, *)$ be a group and $(H, *)$ be a subgroup.
There is one-to-one correspondence between any two right cosets of H in G .

Theorem (Lagrange's theorem)

If $(G, *)$ is a finite group and $(H, *)$ is a subgroup
then order of H is a divisor of order of G .

Proof - $\because G$ is finite, $H \subset G$, H also finite

$$\text{Let } O(G) = n \quad \& \quad O(H) = m$$

Let G is partitioned into finite number of distinct
right cosets of H in G , say k right cosets
namely $H, Ha_1, Ha_2, \dots, Ha_{k-1}$.

Order of each coset is the order of H .

$$\therefore O(Ha_1) = O(Ha_2) = \dots = O(Ha_{k-1}) = O(H) = m$$

$$\text{Also } O(G) = O(H) + O(Ha_1) + \dots + O(Ha_{k-1})$$

$$\begin{aligned} n &= m + m + \dots + m \text{ (k times)} \\ &= km \end{aligned}$$

$\therefore m$ divides n i.e. the order of H divides the
order of G .